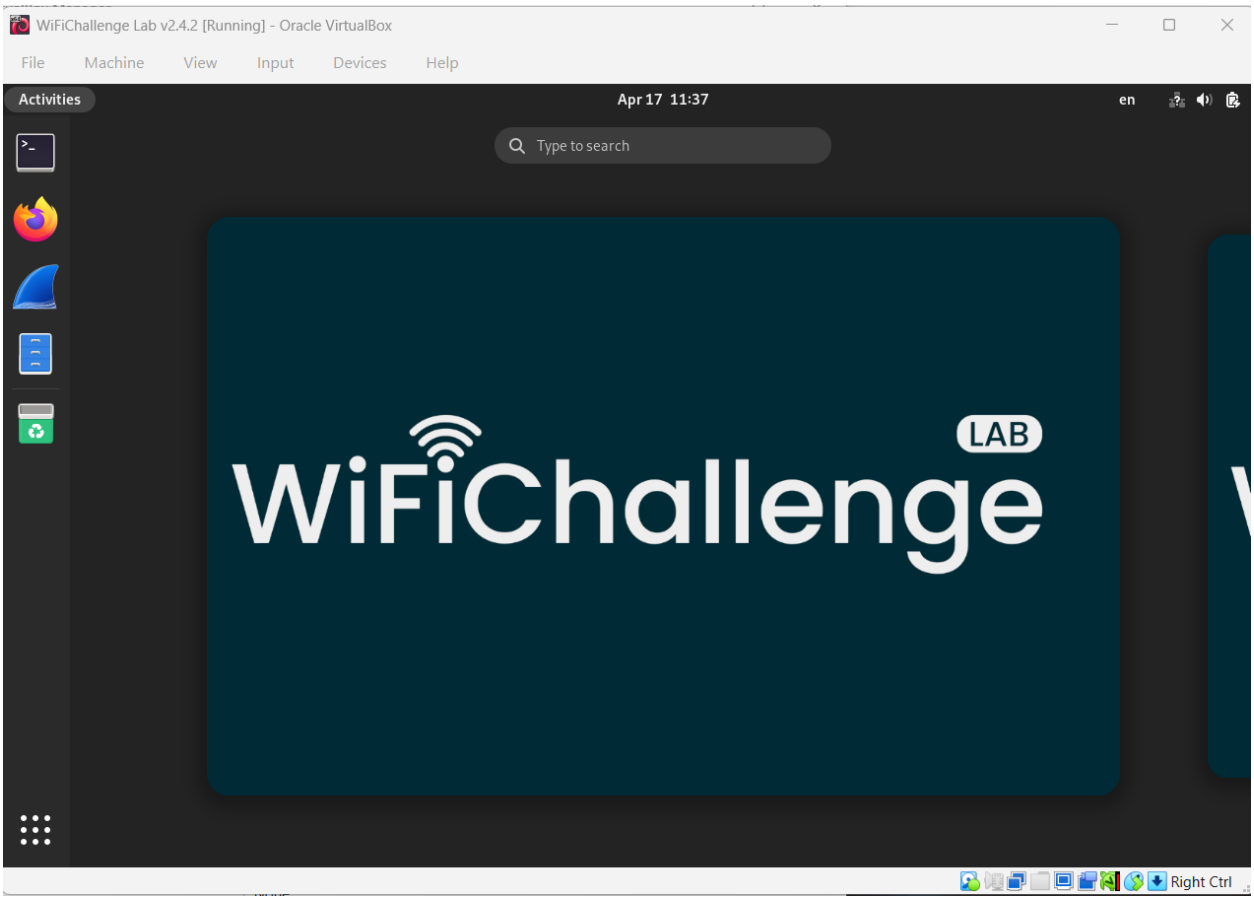
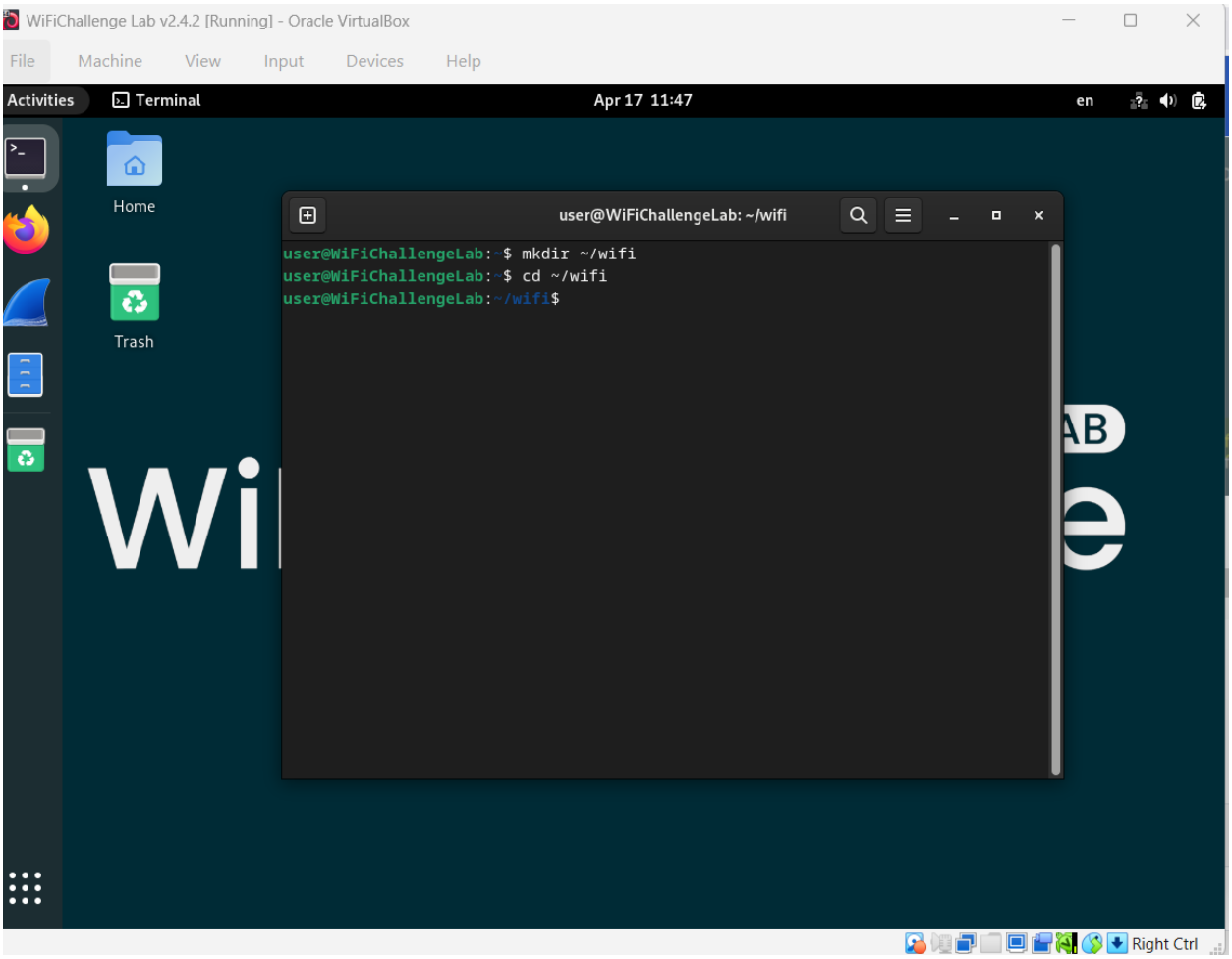
SETTING UP THE WIRELESS LAB

Importing OVA:



Open terminal and create a working folder:



View wireless interface:

```
user@WiFiChallengeLab: ~/wifi

      valid_lft forever preferred_lft forever
: veth97746d8@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue mas
er br-6989dc2e4084 state UP group default
    link/ether 26:99:30:cf:02:90 brd ff:ff:ff:ff:ff:ff link-netnsid 0
: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group
default qlen 1000
    link/ether 32:f4:5f:99:68:ec brd ff:ff:ff:ff:ff:ff permaddr 02:00:00:00:00:0

: wlan1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group
default qlen 1000
    link/ether ee:89:4a:3f:6b:66 brd ff:ff:ff:ff:ff:ff permaddr 02:00:00:00:01:0

0: wlan2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN grou
 default qlen 1000
    link/ether 76:d6:15:d2:bf:77 brd ff:ff:ff:ff:ff:ff permaddr 02:00:00:00:02:0

1: wlan3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN grou
 default qlen 1000
    link/ether fa:65:62:e9:71:b4 brd ff:ff:ff:ff:ff:ff permaddr 02:00:00:00:03:0

2: wlan4: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN grou
 default qlen 1000
    link/ether b2:01:42:70:a4:66 brd ff:ff:ff:ff:ff:ff permaddr 02:00:00:00:04:0
```

Monitor wlan0 interface:

```
user@WiFiChallengeLab: ~/wifi

Found 6 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

    PID Name
    494 avahi-daemon
    510 avahi-daemon
    550 NetworkManager
    565 wpa_supplicant
    853 ifplugd
    989 dhclient

PHY     Interface       Driver          Chipset

phy0    wlan0           mac80211_hwsim  Software simulator of 802.11 radio(s) fo
r mac80211
                (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan
0mon)
                (mac80211 station mode vif disabled for [phy0]wlan0)
phy1    wlan1           mac80211_hwsim  Software simulator of 802.11 radio(s) fo
r mac80211
phy2    wlan2           mac80211_hwsim  Software simulator of 802.11 radio(s) fo
r mac80211
```

What Channel is wifi-global using:

```
user@WiFiChallengeLab: ~/wifi
r --wps --essid
airodump-ng: option '--essid' requires an argument
"airodump-ng --help" for help.
user@WiFiChallengeLab:~/wifi$ sudo airodump-ng wlan0mon --band abg --manufacture
r --wps --essid wifi-global


 CH 169 ][ Elapsed: 1 min ][ 2026-04-17 12:30

 BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH WPS

 F0:9F:C2:71:22:17  -25       26       76    0  44   54e  WPA2 CCMP   MGT  0.0

 BSSID              STATION            PWR    Rate    Lost   Frames  Notes  Pro

 (not associated)   6A:0F:8F:A2:B2:5D  -49    0 - 1      0        1
 (not associated)   64:32:A8:07:6C:40  -26    0 - 1      0        3         AP_r
 (not associated)   64:32:A8:AD:AB:53  -26    0 - 1      0       20         wifi
 (not associated)   64:32:A8:A9:DE:55  -26    0 - 1      0        4         wifi
 (not associated)   F0:9F:C2:71:22:16  -49    0 - 1     98       22         wifi
 (not associated)   78:C1:A7:BF:72:46  -49    0 - 1    147       29         wifi
 (not associated)   B4:99:BA:6F:F9:45  -49    0 - 1     87       27         wifi
 F0:9F:C2:71:22:17  64:32:A8:BA:18:42  -26   54e-54     0       10
 F0:9F:C2:71:22:17  64:32:A8:BC:53:51  -26   54e- 6e    68       88         open
```

```
user@WiFiChallengeLab: ~/wifi

CH 44 ][ Elapsed: 30 s ][ 2026-04-17 12:37 ][ WPA handshake: F0:9F:C2:71:22:17

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

F0:9F:C2:71:AF:2F  -25   0      323      953   17  44   54   WPA3 CCMP   OWE  wifi-gratis
F0:9F:C2:71:22:17  -25   0      323      661   38  44   54e  WPA2 CCMP   MGT  wifi-global
F0:9F:C2:71:22:1A  -25   0      323        8    0  44   54e  WPA2 CCMP   MGT  wifi-corp
F0:9F:C2:71:22:16  -25 100      323        0    0  44   54e  WPA2 CCMP   MGT  wifi-regional
F0:9F:C2:71:22:15  -25 100      323       31    0  44   54e  WPA2 CCMP   MGT  wifi-corp
F0:9F:CB:3F:BC:27  -25 100      323       12    0  44   54   WPA2 CCMP   MGT  wifi-corp-legacy
F0:9F:C2:7A:33:28  -25 100      323       75    0  44   54e  WPA2 CCMP   MGT  wifi-regional-tablets

BSSID              STATION            PWR    Rate    Lost   Frames  Notes  Probes

(not associated)   56:27:C8:63:95:98  -49    0 - 6      0        2
(not associated)   AA:E0:EE:3D:13:B9  -49    0 - 6      0        2
(not associated)   3A:44:40:DE:EC:0C  -49    0 - 6      0        2
(not associated)   16:34:B6:EE:6E:5F  -49    0 - 6      0        2
(not associated)   7E:9B:53:F6:CE:FA  -49    0 - 6      0        2
(not associated)   F0:9F:C2:71:22:16  -49    0 - 6      0       16         wifi-regional
(not associated)   78:C1:A7:BF:72:46  -49    0 - 6    144       24         wifi-offices,Jason
(not associated)   B4:99:BA:6F:F9:45  -49    0 - 6     42       24         wifi-offices,Jason
F0:9F:C2:71:AF:2F  64:32:A8:FA:21:F4  -26   54 -54      0      926
F0:9F:C2:71:22:17  64:32:A8:BA:18:42  -26   54e-54e     0       46
F0:9F:C2:71:22:17  64:32:A8:BC:53:51  -26   54e-54e     0      601  PMKID  open-wifi,home-WiFi,WiFi-Restaurant
F0:9F:C2:71:22:1A  64:32:A8:07:6C:40  -26   24e-54e     0        6
F0:9F:C2:71:22:15  64:32:A8:BA:6C:41  -26   54e-54e     0       30
F0:9F:CB:3F:BC:27  64:32:A8:AD:AB:53  -26    6 - 6     96       48         wifi-corp-legacy
F0:9F:C2:7A:33:28  64:32:A8:BD:64:54  -26   54e-54e     0       57  PMKID
F0:9F:C2:7A:33:28  64:32:A8:A9:DE:55  -26   12e-12e     0       33  PMKID
```

Answer: CH 44

MAC of the wifi-corp client:

64:32:A8:07:6C:40

The Probe station was not found.

Other:

```
user@WiFiChallengeLab: ~/wifi

CH 44 ][ Elapsed: 3 mins ][ 2026-04-18 06:09 ][ WPA handshake: F0:9F:C2:71:22:17

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

F0:9F:C2:71:AF:2F  -25   0     2120     8598   45  44   54   WPA3 CCMP   OWE  wifi-gratis
F0:9F:C2:71:22:1A  -25   0     2120       50    0  44   54e  WPA2 CCMP   MGT  wifi-corp
F0:9F:CB:3F:BC:27  -25 100     2121       60    0  44   54   WPA2 CCMP   MGT  wifi-corp-legacy
F0:9F:C2:7A:33:28  -25   0     2121     2947   15  44   54e  WPA2 CCMP   MGT  wifi-regional-tablets
F0:9F:C2:71:22:17  -25   0     2118     6424    0  44   54e  WPA2 CCMP   MGT  wifi-global
F0:9F:C2:71:22:16  -25 100     2118        0    0  44   54e  WPA2 CCMP   MGT  wifi-regional
F0:9F:C2:71:22:15  -25   0     2120     2898   15  44   54e  WPA2 CCMP   MGT  wifi-corp

BSSID              STATION            PWR    Rate    Lost   Frames  Notes  Probes

(not associated)   F0:9F:C2:71:22:16  -49    0 - 6      0       44         wifi-regional
(not associated)   78:C1:A7:BF:72:46  -49    0 - 6      0      126         wifi-offices,Jason
(not associated)   B4:99:BA:6F:F9:45  -49    0 - 6      0      126         wifi-offices,Jason
F0:9F:C2:71:AF:2F  64:32:A8:FA:21:F4  -26   54 -54      0     8434
F0:9F:C2:71:22:1A  64:32:A8:07:6C:40  -26   12e-12e     0       84  PMKID  AP_router,wifi-corp
F0:9F:CB:3F:BC:27  64:32:A8:AD:AB:53  -26    6 - 1      0      258         wifi-corp-legacy
F0:9F:C2:7A:33:28  64:32:A8:A9:DE:55  -26   48e-54e     0       53  PMKID  wifi-regional-tablets
F0:9F:C2:7A:33:28  64:32:A8:BD:64:54  -26   54e-54e     0     2914  PMKID  wifi-regional-tablets
F0:9F:C2:71:22:17  64:32:A8:BA:18:42  -26   24e-24e     0      270  PMKID
F0:9F:C2:71:22:17  64:32:A8:BC:53:51  -26   54e- 6e     0     6081  PMKID  open-wifi,home-WiFi,WiFi-Restaurant
F0:9F:C2:71:22:15  64:32:A8:BA:6C:41  -26   54e-54e     0     2906  PMKID  wifi-corp
```